

Semester: IV
Course Title: Object Oriented Programming Laboratory
Academic Year: 2025-26

Course Code: 22UCSL403
Division: A & B
Course Instructors: Prof. IRU, Prof. CC

<u>Part-A</u>	
1)	A vehicle is identified by how many wheels it has. A car, bike and truck are identified by their engine number, chassis number and manufacturer. Car <u>is-a</u> vehicle, Bike <u>is-a</u> vehicle and Truck <u>is-a</u> vehicle. Using this information, write a Java Program that displays all the information of vehicle, car, bike and truck. Also draw the class diagram for the problem scenario.
2)	Write a Java Program that takes a string as command-line argument and checks whether the string is palindrome or not. If the string is palindrome, then print proper message; otherwise, generate a user-defined exception StringNotPalindromeException .
3)	A chemical company named “XYZ Chemicals Pvt. Ltd.” manufactures various chemicals. To automate and monitor the manufacturing process, they are planning to install an “ Automated Manufacturing and Maintenance System ”. One important task of this system is to sense the rise in temperature of the furnace and monitor its temperature. If the temperature of the furnace rises above 300 °C, then it should immediately generate an alarm. It should also generate a separate alarm if the temperature of the furnace falls below 100 °C. You are hired as a Java Programmer to develop this system. Write a Java Program which simulates the above scenario using exception handling mechanism.
4)	Write a Java program that provides a generic sort method, which can sort elements of any reference type.
5)	Postgraduate course of the CSE department can accommodate a maximum of 20 students. Each PG student in the department is identified by his/her Roll No, USN, Name, Semester and Mobile Number. Using appropriate Collection class, write a Java Program to simulate the following scenarios: i) On request, the capacity was increased from 20 to 25 ii) Only 3 students enroll for the course iii) Two students bearing Roll Numbers 2 and 3 voluntarily unroll from the course

	iv) After a month, two students get enrolled in the course v) Print the information of all the students currently enrolled in the course																		
6)	Write a JSP program that implements a simple calculator. The program must provide the features to perform addition, subtraction, multiplication and division of two numbers based on user's choice.																		
7)	Write a JSP program that reads marks scored by a student in a course (out of 100) and displays the grade obtained. Use the following table to compute the grades: <table><tr><th>Marks (out of 100)</th><th>(90 – 100)</th><th>(80 – 89)</th><th>(70 – 79)</th><th>(60 – 69)</th><th>(50 – 59)</th><th>(45 – 49)</th><th>(40 – 44)</th><th>(39 – 0)</th></tr><tr><th>Grade</th><td>O</td><td>A+</td><td>A</td><td>B+</td><td>B</td><td>C</td><td>P</td><td>F</td></tr></table>	Marks (out of 100)	(90 – 100)	(80 – 89)	(70 – 79)	(60 – 69)	(50 – 59)	(45 – 49)	(40 – 44)	(39 – 0)	Grade	O	A+	A	B+	B	C	P	F
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Grade	O	A+	A	B+	B	C	P	F											
8)	Write a Java program that uses streams to copy the contents of one file into another. The program must take the file names from command-line arguments and perform error handling if the entered file is invalid or file cannot be opened.																		
<u>Part-B</u>																			
1)	Write a Java program that uses threads to compute multiplication of two matrices. The program should perform multiplication of matrices of arbitrary order. Also, proper error handling mechanism should be used. Methodology: Matrix multiplication is implemented using the formula: $C_{ij} = A_{ik} * B_{kj}$. This formula is used for generating each element of the final matrix. The multiplication must be performed by separate threads. Ex: If order of final matrix is 3X3, then total of 9 threads should be created; each thread computing individual elements of the final matrix.																		
2)	Write a Java program that creates three threads. The first thread prints prime numbers from 1 to 100; second thread prints prime numbers from 101 to 200 and third thread prints prime numbers from 201 to 300. Note: The three threads must call the same method generatePrime() to print the prime numbers.																		
3)	Write a Java program that simulates client-server interaction using threads. Methodology: Two threads should be created; one thread should act as server, and the other one should act as client. The server thread should accept a string from the client																		

	thread. Upon receiving and displaying the string, the server thread should send back the acknowledgement message “ Message Received ” to the client thread.										
4)	Student in SDMCET is identified by USN, first name, last name, branch and division. Write a Java program using generics to provide facility to sort the students based on USN, first name and division. The sort option is to be read by the user. Note: The program must implement compareTo() method available in Comparable<T> interface to sort the students based on the user’s choice.										
5)	Write a Java program that provides the following implementations: i) <u>HTML file</u>: displays a text box to read a line of text and provides a submit button ii) <u>Java Servlet</u>: provides features to handle the request (from HTML file), processes the request and sends the number of vowels and consonants present in the line of text in HTML format										
6)	Write a Java servlet that implements a simple calculator. The program must provide following implementations: i) <u>HTML file</u>: displays a form, that reads two numbers from the user and provides the buttons: Add, Subtract, Multiply and Divide ii) <u>Java Servlet</u>: provides features to handle the request (from HTML file), computes the result and sends back the result in HTML format										
7)	Write a Java program to implement the following scenario: Consider the following table available in the MySQL instance of StudentDB: <table border="1"><thead><tr><th colspan="2">StudentInfo</th></tr></thead><tbody><tr><td>RollNo</td><td>Int</td></tr><tr><td>Name</td><td>varchar(20)</td></tr><tr><td>USN</td><td>varchar(10)</td></tr><tr><td>Division</td><td>varchar(1)</td></tr></tbody></table> Write a Java program that displays contents of the above-mentioned table on the console.	StudentInfo		RollNo	Int	Name	varchar(20)	USN	varchar(10)	Division	varchar(1)
StudentInfo											
RollNo	Int										
Name	varchar(20)										
USN	varchar(10)										
Division	varchar(1)										

8)	Write a Java program that uses connection-oriented sockets and provides multi-way communication between client and server as shown below: Server: Prompts client to enter a string Client: <provides a string> Server: Reverses the string and displays the reversed string and length of the string
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